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Dressing design incorporating formed 3D textile for the delivery of therapeutic negative pressure and compressive forces to a treatment site

Abstract

A system for applying negative pressure to a joint positioned in a treatment area. The system includes a negative pressure dressing. The negative pressure dressing includes a compressive layer and a sealing layer. The compressive layer includes a first surface and a second, treatment area-facing. The compressive layer further includes a first elongated portion configured to be positioned proximate the joint. The first elongated portion includes a first end and a second end. The compressive layer further includes a second elongated portion spaced from the first elongated portion and configured to be positioned proximate the joint. The second elongated portion includes a first end and a second end. The compressive layer further includes an interconnecting portion extending between the first elongated portion and the second elongated portion. The interconnecting portion is configured to overlie at least a portion of the joint. The compressive layer further includes a plurality of channels formed in the first surface and extending proximate the second surface. The plurality of channels is formed in at least one of the first end and the second end of the first elongated portion and at least one of the first end and the second end of the second elongated portion. The sealing layer is configured to form a seal around a perimeter of the negative pressure dressing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/081,261, filed on Oct. 27, 2020, which claims the benefit of priority to U.S. Provisional Application No. 62/929,197, filed on Nov. 1, 2019, which is incorporated herein by reference in its entirety.

BACKGROUND
(1) The present disclosure relates generally to a negative pressure therapy (NPT) system, and more particularly, to a NPT system adapted to provide NPT to treat joint injuries.
(2) Negative pressure wound therapy (NPWT) is a type of wound treatment therapy that involves applying a negative pressure to a wound site to promote wound. NPWT applies negative pressure to the wound area to drain fluids from the wound area as the wound heals. NPWT can be used to treat a variety of surgical wounds.
SUMMARY
(3) One implementation of the present disclosure is system for applying negative pressure to a joint positioned in a treatment area. The system includes a negative pressure dressing including a compressive layer and a sealing layer. The compressive layer includes a first surface and a second, treatment area-facing surface. The compressive layer further includes a first elongated portion configured to be positioned proximate the joint. The first elongated portion includes a first end and a second end. The compressive layer further includes a second elongated portion spaced from the first elongated portion and configured to be positioned proximate the joint. The second elongated portion including a first end and a second end. The compressive layer further includes an interconnecting portion extending between the first elongated portion and the second elongated portion. The interconnecting portion is configured to overlie at least a portion of the joint. The compressive layer further includes a plurality of channels formed in the first surface and extending proximate the second surface. The plurality of channels is formed in at least one of the first end and the second end of the first elongated portion and at least one of the first end and the second end of the second elongated portion. The sealing layer is configured to form a seal around a perimeter of the negative pressure dressing.
(4) Another implementation of the present disclosure is a system for applying a negative pressure to a joint positioned in a treatment area. The system includes a negative pressure dressing including a compressive layer and a sealing layer. The compressive layer includes a first surface and a second, treatment area-facing surface. The compressive layer further includes a first layer having first material properties. At least a portion of the first layer forms the second surface of the compressive layer. The compressive layer further includes a plurality of ribs formed on the first layer. The plurality of ribs has second material properties. The plurality of ribs is spaced apart by a plurality of channels. The plurality of ribs form the first surface of the compressive layer. The sealing layer overlies the first layer and the second layer. The compressive layer is configured to collapse in a substantially vertical direction and then collapse in a substantially horizontal direction upon application of the negative pressure.
(5) Another implementation of the present disclosure is a system for applying negative pressure to a

joint positioned in a treatment area. The system includes a negative pressure dressing and a negative pressure source. The negative pressure dressing includes a compressive layer and a sealing layer. The compressive layer includes a first surface and a second, treatment area-facing surface. The compressive layer is configured for compression in at least a substantially vertical direction and a substantially horizontally direction. The compression in the substantially horizontal direction is larger than the compression in the substantially vertical direction. The compressive layer further includes a first portion including the second surface and a second portion vertically spaced from the first portion and substantially parallel to the first portion. The sealing layer is configured to form a seal around a perimeter of the negative pressure dressing. The negative pressure source is in fluid communication with the negative pressure dressing and configured to apply a negative pressure to the compressive layer to compress at least the compressive layer.

(6) Another implementation of the present disclosure is a compressive layer for a negative pressure treatment dressing. The compressive layer includes a first surface and a second, treatment area-facing surface. The compressive layer further includes a first portion and a second portion. The first portion includes the second surface of the compressive layer. The first portion has a first compression modulus in a substantially horizontal direction and a second compression modulus in a substantially vertical direction. The first compression modulus is smaller than the second compression modulus. The second portion extends from the first portion and includes the first surface of the compressive layer. The second portion has a first compression modulus in a substantially horizontal direction. The first compression modulus of the second portion is different than the first compression modulus of the first portion.

(7) Another implementation of the present disclosure is a system for applying negative pressure to a joint positioned in a treatment area. The system includes a negative pressure dressing and a negative pressure source. The negative pressure dressing includes a compressive layer and a sealing layer. The compressive layer includes a first surface and a second, treatment area-facing surface. The compressive layer further includes a first portion including the second surface of the compressive layer. The first portion has a first compression modulus in a direction substantially parallel to the treatment area and a first thickness substantially perpendicular to the treatment area. The compressive layer further includes a second portion including the first surface of the compressive layer. The second portion has a first compression modulus in the direction substantially parallel to the treatment area and a second thickness in the direction substantially perpendicular to the treatment area. The second thickness is smaller than the first thickness and the second compression modulus is different than the first compression modulus. The sealing layer is configured to overlie the compressive layer and to form a seal around a perimeter of the compressive layer. The negative pressure source is in fluid communication with the negative pressure dressing and configured to apply a negative pressure to the compressive layer to compress at least the compressive layer.

(8) Those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices and/or processes described herein, as defined solely by the claims, will become apparent in the detailed description set forth herein and taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. **1** is a front view of a negative pressure dressing according to an exemplary embodiment.

(2) FIG. **2** is a side view of the negative pressure dressing of FIG. **1** according to an exemplary embodiment.

(3) FIG. **3** is an exploded view of the negative pressure dressing of FIG. **1** according to an

exemplary embodiment.
(4) FIG. **4** is a front view of a compressive layer of a negative pressure dressing for use with the negative pressure dressing of FIG. **1** according to some embodiments.
(5) FIG. **5** is a perspective view of a three-dimensional textile for use with the negative pressure dressing of FIG. **1** according to an exemplary embodiment.
(6) FIG. **6** is a perspective view of a welded foam material for use with the negative pressure dressing of FIG. **1** according to another exemplary embodiment.
(7) FIG. **7** is a side view of the negative pressure dressing of FIG. **1** secured to a patient according to an exemplary embodiment.
(8) FIG. **8** is a front view of the negative pressure dressing of FIG. **1** secured to a patient according to an exemplary embodiment.

DETAILED DESCRIPTION

Overview

(9) Referring generally to the FIGURES, a negative pressure dressing for use with a negative pressure system for treating injuries to joints is shown, according to various embodiments. More specifically, the negative pressure dressing is primarily intended for treating injuries to joints, such as sprains, or joint conditions such as arthritis, but may be used to treat other injuries under appropriate circumstances. The negative pressure dressing is described herein in the context of treating a knee joint, but the negative pressure dressing can be used to treat other joints, such as ankle joints, hip joints, wrist joints, elbow joints, and shoulder joints.
(10) The phrase “negative pressure” means a pressure less than an ambient or atmospheric pressure. While the amount and nature of reduced pressure applied to the treatment site can vary according to the application, the reduced pressure typically is between −5 mm Hg and −500 mm Hg and more typically between −100 mm Hg and −300 mm Hg. The negative pressure dressing described herein is shaped to overlie the treatment area, more specifically, proximal portion of the body surface adjacent the joint, the joint, and a distal portion of the body surface adjacent the joint. The negative pressure dressing includes a first elongated portion configured to engage the proximal portion of the body surface, a connecting portion configured to engage the joint, and a second elongated portion configured to engage the distal portion of the body surface.
(11) The negative pressure dressing includes a compressive layer. The compressive layer is formed of a material that collapses laterally and vertically under negative pressure. The compressive layer includes a first elongated portion, connecting portion, and a second elongated portion. A plurality of channels that form a plurality of ribs is formed on ends of the first elongated portion and ends of the second elongated portion to facilitate generally lateral compression under negative pressure. The phrase “channel” means an elongated open top recess that extends longitudinally between the ribs.
(12) When the negative pressure dressing is under negative pressure, the compressive layer first collapses in a generally vertical direction. The compressive layer then collapses in a generally lateral generation, causing bending and puckering of the compressive layer. This bending generates upward lifting forces about substantially a perimeter of the treatment area. The upward lifting forces improve circulation of blood and lymph fluids in the treatment area, reducing swelling and/or inflammation of the treatment site.
(13) Additional features and advantages of the negative pressure therapy system are described in detail below.

Negative Pressure Therapy System

(14) Referring to FIGS. **1-3**, a negative pressure dressing **10** for use with a negative pressure therapy (NPT) system **12** (FIGS. **7-8**) is shown, according to an exemplary embodiment. FIG. **1** is a front view of the negative pressure dressing **10**. FIG. **2** is a side view of the negative pressure dressing **10**. FIG. **3** is an exploded view of the negative pressure dressing **10**. In various embodiments, the negative pressure dressing **10** can be used to treat joint injuries and/or other

painful joint conditions. In various embodiments, the negative pressure dressing **10** is applied to intact skin and can be used to treat joints such as knees, ankles, shoulders, elbows, wrists, and hips. The negative pressure dressing **10** is configured to apply a combination of vertical and lateral compression to generate lifting forces within a perimeter of the treatment site. The lifting forces improve circulation of blood and lymph fluid, thereby reducing swelling and/or inflammation proximate the treatment site **14**.

(15) In various embodiments, the negative pressure dressing **10** can be formed as a substantially flat sheet. The negative pressure dressing **10** is adapted to conform to treatment sites **14** having high curvature and mobility, such as knee and elbow joints. The treatment site **14** includes the joint, a surface of the body distal to and adjacent the joint, and a surface of the body proximal to and adjacent the joint. The word "distal" is generally used herein to refer to a portion of the body that is away from a center of the body or an attachment point between the body and the limb including. The word "proximal" is generally used herein to refer to a portion of the body that is near a center of the body or is near an attachment point between the body and the limb including the joint.

(16) The negative pressure dressing **10** is adapted to move with the patient as the patient moves the treated joint. The negative pressure dressing **10** includes a first elongated portion **18**, a second elongated portion **22**, and a connecting portion **26**. The first elongated portion **18** and the second elongated portion **22** are generally parallel. The first elongated portion **18** and the second elongated portion **22** are configured to overlie body surfaces adjacent the joint. In the illustrated embodiment, the first elongated portion **18** is longer than the second elongated portion **22**. Accordingly, the first elongated portion **18** can overlie a proximal body surface adjacent the joint and the second elongated portion can overlie a distal body surface adjacent the joint since the distal body surface is generally larger than the proximal surface. The connecting portion **26** is configured to overlie the joint. Although the negative pressure dressing **10** is described herein in the context of a knee joint, the negative pressure dressing **10** can also be used to treat ankle joints, hip joints, elbow joints, wrist joints, and shoulder joints.

(17) The negative pressure dressing **10** is shown to include a plurality of layers, including a sealing layer **30**, a compressive layer **34**, and an optional non-adherent layer **38**. The negative pressure dressing **10** includes a longitudinal axis **42** defining a longitudinal direction, a lateral axis **46** defining a lateral direction, and a vertical axis **50** defining a vertical direction. The negative pressure dressing **10** is symmetric about the longitudinal axis **42**.

Sealing Layer

(18) With continued reference to FIGS. **1-3**, the sealing layer **30** is shown to include a first surface **52** and a second, treatment area-facing surface **54** opposite the first surface **52**. When the negative pressure dressing **10** is applied to a treatment area, the first surface **52** faces away from the treatment area, whereas the second surface **54** faces toward the treatment area. The sealing layer **30** supports the compressive layer **34** and the non-adherent layer **38** and provides a barrier to passage of microorganisms through the negative pressure dressing **10**. In some embodiments, the sealing layer **30** is an elastomeric material or may be any material that provides a fluid seal. "Fluid seal" means a seal adequate to hold pressure at a desired site given the particular reduced-pressure subsystem involved. The term "elastomeric" means having the properties of an elastomer and generally refers to a polymeric material that has rubber-like properties. Examples of elastomers may include, but are not limited to, natural rubbers, polyisoprene, styrene butadiene rubber, chloroprene rubber, polybutadiene, nitrile rubber, butyl rubber, ethylene propylene rubber, ethylene propylene diene monomer, chlorosulfonated polyethylene, polysulfide rubber, polyurethane, EVA film, co-polyester, and silicones. As non-limiting examples, the sealing layer **30** may be formed from materials that include a silicone, 3M Tegaderm® drape material, acrylic drape material such as one available from Avery, or an incise drape material.

(19) In the illustrated embodiment, the sealing layer **30** defines a cavity **56** (FIG. **3**) for receiving the compressive layer **34** and the non-adherent layer **38**. As shown in FIG. **3**, the compressive layer

34 and the non-adherent layer **38** can have a similar perimeter or profile. As is shown in FIGS. **1** and **3**, the first surface **52** of the sealing layer **30** includes a negative pressure port **58** for fluid communication with the NPT system **12** as shown in FIGS. **7-8**. In some embodiments, a perimeter of the sealing layer **30** extends beyond (e.g. circumscribes) the perimeter of the compressive layer **34** to provide a margin **62**. The second surface **54** of the margin is coated with an adhesive, such as an acrylic adhesive, a silicone adhesive, and/or other adhesives. The adhesive applied to the second surface **54** of the margin **62** is intended to ensure that the negative pressure dressing **10** adheres to the surface of the patient's skin (as shown in FIGS. **7-8**) and that the negative pressure dressing **10** remains in place throughout the wear time. The margin **62** may extend around all sides of the compressive layer **34** such that the negative pressure dressing **10** is a so-called island dressing. In other embodiments, the margin **62** and can be eliminated and the negative pressure dressing **10** can be adhered to the surface using other techniques.
(20) As shown in FIGS. **1-3**, the first surface **52** of the sealing layer **30** further includes a plurality of articulations **66** proximate the connecting portion **26**. The plurality of articulations **66** are positioned to generally overlie the joint and allow more bending and straightening of the joint when the negative pressure dressing **10** is secured to the patient.
The Compressive Layer
(21) Referring to FIG. **3**, the compressive layer **34** is shown to include a first surface **70** and a second, treatment area-facing surface **74** opposite the first surface **70**. When the negative pressure dressing **10** is applied to a treatment site **14**, the first surface **70** faces away from the treatment site **14**, and the second surface **74** faces toward the treatment site **14**. In some embodiments, the first surface **70** of the compressive layer **34** contacts the second surface **74** of the sealing layer **30**. In some embodiments, the second surface **74** of the compressive layer **34** contacts a patient's skin. In embodiments that include the optional non-adherent layer **38**, the second surface **74** of the compressive layer **34** contacts the non-adherent layer **38**. The compressive layer **34** is adapted to compress under negative pressure. More specifically, the compressive layer **34** is adapted to first collapse in a generally vertical direction defined by the vertical axis **50**. The compressive layer **34** is then adapted to collapse in a generally horizontal (e.g., lateral) direction defined by the lateral axis **46** towards the longitudinal axis **42** of the negative pressure dressing **10** and bend upwards, generating lift. This combination of lateral compression and improves circulation and flow of lymph fluids throughout the treatment area, thereby reducing swelling and/or inflammation in the treatment area.
(22) As is best shown in FIG. **4**, the compressive layer **34** is generally symmetrical about a longitudinal axis **42** and includes a first elongated portion **78**, a second elongated portion **82**, and a connecting portion **86** extending between the first elongated portion **78** and the second elongated portion **82**. The compressive layer **34** can have a width W that is between approximately 80 mm and approximately 300 mm. The compressive layer **34** can have a thickness T that is between approximately 2 mm and approximately 8 mm. The first elongated portion **78** can have a length L.sub.1 that is between approximately 50 mm and approximately 100 mm. The first elongated portion **78** can have a width W.sub.1 that is between approximately 40 mm and approximately 80 mm. The second elongated portion can have a length L.sub.2 that is between approximately 40 mm and approximately 80 mm. The second elongated portion can have a width W.sub.2 that is between approximately 20 mm and approximately 60 mm. The connecting portion **86** can have a length L.sub.c that is between approximately 30 mm and approximately 60 mm. The connecting portion **86** can have a width W.sub.c that is between approximately 30 mm and approximately 50 mm. In the illustrated embodiment, the compressive layer **34** is symmetric about the longitudinal axis **42**.
(23) The first elongated portion **78** has a first end **90** and a second end **94**. The first elongated portion **78** can be wrapped about a first body surface near the joint. More specifically, the first elongated portion **78** is intended to be secured about a proximal body surface proximate the joint. The second elongated portion has a first end **98** and a second end **102**. The second elongated

portion **22** can be wrapped about a second body surface proximate the joint. As is seen in FIGS. **6-7**, the first elongated portion **78** and the second elongated portion **82** are positioned on opposing sides of the joint. A central portion **106** extends between the first end **90** and the second end **94** of the first elongated portion **78**, the first end **102** and the second end **102** of the second elongated portion **82**, and along the connecting portion **86**.

(24) As is best shown in FIGS. **2-4**, a plurality of ribs **110** is formed in the first end **90** of the first elongated portion **78**, the second end **94** of the first elongated portion **78**, the first end **98** of the second elongated portion **82**, and the second end **102** of the second elongated portion **82**. As shown in FIGS. **2** and **3**, a plurality of channels **114** is formed in the first surface **70** of the compressive layer **34** and extend towards the second surface of the compressive layer **34**. The plurality of channels **114** form a plurality of ribs **110** between adjacent channels of the plurality of channels **114**. The plurality of channels **114** does not extend through the second surface **74**. The plurality of channels **114** are elongated open top recesses extending longitudinally between the ribs. The plurality of ribs **110** and the plurality of channels **114** are configured to facilitate lateral compression (e.g., compression towards the longitudinal axis **42** in the lateral direction defined by the lateral axis **46**) of the compressive layer **34** by reducing an amount of material present in the first end **90** and second end **94** of the first elongated portion **78** and the first end **98** and the second end **102** of the second elongated portion **82**.

(25) In some embodiments, the plurality of channels **114** have a depth D that is between approximately 0.5 cm and approximately 2 cm. In the illustrated embodiment, each of the ribs of the plurality of ribs **110** has a width W.sub.r of approximately 5 mm. In the illustrated embodiment, a spacing (e.g., channel width) S.sub.r between adjacent ribs is approximately 3 mm-4 mm. In other implementations, other dimensions of the depth D, the width W.sub.r, and the spacing S.sub.r can be used. For example, in some embodiments, a spacing between the ribs of the plurality of ribs can be graduated to facilitate lateral compression, such that the spacing S.sub.r between adjacent ribs of the plurality of ribs **110** increases in a laterally inward direction. In some embodiments, the width W.sub.r of the ribs the plurality of ribs **110** can be graduated to facilitate lateral compression, such that widths W.sub.r of adjacent ribs of the plurality of ribs **110** decreases in a laterally inward direction. As is described in greater detail below, in some embodiments, the compressive layer **34** can be made of a textile material and the plurality of ribs **110** and the plurality of channels can be knit into the compressive layer **34** by varying the kinit pattern of the textile material. In some embodiments, the compressive layer can be made of a foam material and the ribs can be welded into or cut into the compressive layer **34**.

(26) As shown in FIG. **5**, in some embodiments, the compressive layer **34** can be formed of a three-dimensional (3D) textile material **118** having alternate thick **122** and thin regions **126**. One such textile material is Balltex **3520** that is available from W. Ball & Sons, Ltd. Any material or combination of materials can be used for the compressive layer **34** provided that the compressive layer **34** is operable to distribute the reduced pressure and provide a distributed compressive force at the treatment site **14**. The textile material **118** can include channels for distributing pressure and/or fluids knit into the textile material **118**.

(27) The Balltex **3520** material is a 100% polyester material that has a weight of approximately 300 g/m.sup.2—approximately 340 g/m.sup.2. In some embodiments, at least a portion of the material may be coated to make the at least a portion of the material more or less hydrophilic. An exemplary coating technique is Plasma Coating by P**2***i* Ltd.

(28) As is shown in the inset of FIG. **5**, the three-dimensional textile material **118** includes a first portion **130**, a interconnecting portion **134**, and a second portion **138**. In the illustrated embodiment, the first portion **130** forms the second surface **74** of the compressive layer **34**, the second portion **138** forms the first portion **130** of the compressive layer **34**, and the interconnecting portion **134** that extends between the first portion **130** and the interconnecting portion **134** to secure the first portion **130** and the interconnecting portion **134** together. The first portion **130** and the

second portion **138** are substantially parallel. The interconnecting portion **134** is substantially perpendicular to the first portion **130** and the second portion **138**. The first portion **130** is a continuous layer configured to abut the patient's skin or the optional non-adherent layer **38**.

(29) The textile material **118** is configured to have a vertical compression modulus in a generally vertical direction defined by the vertical axis **50** and a lateral compression modulus in a generally lateral direction defined by the lateral axis **46**. In the illustrated embodiment, lateral compression modulus is lower than the vertical compression modulus to facilitate lateral compression. The first portion **130**, the interconnecting portion **134**, and the second portion **138** can be non-interdependent materials, such that the first portion **130**, the interconnecting portion **134**, and the second portion **138** can be made of different materials. In some embodiments, the first portion **130**, the interconnecting portion **134**, and the second portion **138** can have different material properties. For example, in some embodiments, the first portion **130**, the interconnecting portion **134**, and the second portion **138** can have different textile knit patterns, different weights, different densities, different fibers, and/or different stiffnesses. For example, in some embodiments, the first portion **130** can have a first textile knit pattern including a first plurality of pores (e.g., spaces between textile fibers), the interconnecting portion **134** can have a second textile knit pattern including a second plurality of pores, and the second portion **138** can have a third textile knit pattern including a third plurality of pores. The first knit pattern can have smaller voids (e.g., be a tighter knit) than the third knit pattern and/or the second knit pattern. In some embodiments, the second knit pattern and/or the third knit pattern can have larger voids (e.g., be a looser knit) to reduce a pressure drop across the compressive layer **34**. In other embodiments, the first portion **130**, the interconnecting portion **134**, and the second portion **138** can be made of different materials. In some embodiments, the first portion **130**, the interconnecting portion **134**, and the second portion **138** can be treated with different materials. For example, in some embodiments, the first portion **130** can be treated to prevent the compressive layer **34** from irritating a patient's skin. The materials and the material properties of the first portion **130**, the interconnecting portion **134**, and the second portion **138** and the dimensions of the plurality of ribs **110** and the plurality of channels **114** can be varied in different embodiments and/or different applications to customize an amount of lateral compression and lift generated by the compressive layer **34** to a particular joint.

(30) With continued reference to FIG. **5**, the three dimensional textile material **118** includes thick regions **122** and thin regions **126**. The thick regions **122** form the plurality of ribs **110** and the thin regions **126** form the plurality of channels **114** positioned adjacent the plurality of ribs **110**. In some embodiments, the thick region **122** can be formed by the first portion **130**, the interconnecting portion **134**, and the second portion **138**. The thin region **126** can be formed by at least the first portion **130** and can include a portion of the interconnecting portion **134** and/or a portion of the second portion **138**. In some embodiments, such as the embodiment shown in FIG. **5**, the entire three dimensional textile material **118** can include the thick regions **122** and thin regions **126**. In other embodiments, such as the embodiment shown in FIGS. **1-4**, the textile material **118** can include the thick region **122** on the central portion **106**. The thick region **122** can be formed of the first portion **130**, the interconnecting portion **134**, and the second portion **138**. The textile material **118** can further include the plurality of ribs **110** and the plurality of channels **114** (e.g., the thick regions **122** and the thin regions **126**, respectively) proximate the ends **90**, **94** of the first elongated portion **78** and the ends **98**, **102** of the second elongated portion **82**. Such a configuration facilitates a combination of lateral compression and upward bending proximate the ends **90**, **94** of the first elongated portion **78** and the ends **98**, **102** of the second elongated portion **82**, which generates a generally circumferential lifting force around the joint. The lifting force around the joint facilitates circulation and/or lymph flow in the treatment site, reducing swelling and/or inflammation of the treatment site.

(31) As shown in FIG. **6**, in some embodiments, the compressive layer **34** can be made from a porous and permeable foam-like material **146** and, more particularly, a reticulated, open-cell

polyurethane or polyether foam that allows good permeability of while under a reduced pressure. One such foam material that has been used is the VAC® Granufoam® material that is available from Kinetic Concepts, Inc. (KCI) of San Antonio, Tex. Any material or combination of materials might be used for the compressive layer **34** provided that the compressive layer **34** is operable to distribute the reduced pressure and provide a distributed compressive force along the treatment site.
(32) The reticulated pores of the Granufoam® material **146** that are in the range from about 400 to 600 microns, are preferred, but other materials may be used. The compression modulus of Granufoam at 65% compression is approximately 3 kPa (0.43 psi). The density of the absorbent layer material, e.g., Granufoam® material, is typically in the range of about 1.3 lb/ft.sup.3-1.6 lb/ft.sup.3 (20.8 kg/m.sup.3-25.6 kg/m.sup.3). A material with a higher density (smaller pore size) than Granufoam® material may be desirable in some situations. For example, the Granufoam® material or similar material with a density greater than 1.6 lb/ft.sup.3 (25.6 kg/m.sup.3) may be used. As another example, the Granufoam® material or similar material with a density greater than 2.0 lb/ft.sup.3 (32 kg/m.sup.3) or 5.0 lb/ft.sup.3 (80.1 kg/m.sup.3) or even more may be used. The more dense the material is, the higher compressive force that may be generated for a given reduced pressure. If a foam with a density less than the tissue at the tissue site is used as the absorbent layer material, a lifting force may be developed. In one illustrative embodiment, a portion, e.g., the edges, of the negative pressure dressing **10** may exert a compressive force while another portion, e.g., the central portion **106**, may provide a lifting force.
(33) Among the many possible compressive layer **34** materials, the following may be used: Granufoam® material or a Foamex® technical foam (www.foamex.com). In some instances it may be desirable to add ionic silver to the foam in a microbonding process or to add other substances to the absorbent layer material such as antimicrobial agents. The absorbent layer material may be isotropic or anisotropic depending on the exact orientation of the compressive forces that are desired during the application of reduced pressure. The compressive layer **34** material may also be a bio-absorbable material.
(34) As shown in FIG. **6**, the foam material **146** can be a welded foam material **146** including a plurality of welds **150**. The plurality of welds **150** form alternately spaced thick regions (e.g., the foam material **146** extending between the plurality of welds **150**) **154** and thin regions **158** defined by the plurality of welds **150**. The foam material **146** is configured to have a vertical compression modulus in a generally vertical direction defined by the vertical axis **50** and a lateral compression modulus in a generally lateral direction defined by the lateral axis **46**. In the illustrated embodiment, lateral compression modulus is lower than the vertical compression modulus to facilitate lateral compression. The thick regions **154** and the thin regions **158** can have different material properties. For example, in some embodiments, the thin regions **158** and the thick regions **154** can have different densities, and/or different stiffnesses. For example, in the illustrated embodiments, the thin regions **158** are denser than the thick regions **154**. The thin regions **158** can therefore generate larger compressive forces in the generally lateral direction defined by the lateral axis **46** than the thick regions **154**. The positioning of the plurality of welds **150** (e.g., spacing and number of welds) can be varied in different embodiments and/or different applications to customize an amount of lateral compression and lift generated by the compressive layer **34** to a particular joint.
(35) FIG. **6** illustrates an exemplary foam compressive layer material **146** including the thick regions **154** and the thin regions **158**. The thick regions **154** form the plurality of ribs **110** and the thin regions **158** are formed the plurality of welds **150**. In the embodiment illustrated in FIG. **6**, the plurality of welds **150** forms the plurality of channels **114**. In some embodiments, such as the embodiment shown in FIG. **6**, the entire three dimensional textile material includes thick regions **154** and thin regions **158**. The spacing of the welds of the plurality of welds **158** varies such that the central portion **106** of the foam material **146** is unwelded and a portion of the foam material **146** includes the plurality of welds **150**. In embodiments of the negative pressure dressing **10**, such as

the embodiment shown in FIGS. **1-4**, the foam material **146** can include an unwelded portion proximate the central portion **106** the negative pressure dressing **10**, and include welded portions proximate the ends **90**, **94** of the first elongated portion **18** and the ends **98**, **102** of the second elongated portion **22**. Such a configuration facilitates a combination of lateral compression and upward bending proximate the ends **90**, **94** of the first elongated portion **18** and the ends **90**, **94** of the second elongated portion **22**, which generates a generally circumferential lifting force around the joint. The lifting force around the joint facilitates circulation and/or lymph flow in the treatment site, thereby reducing swelling and/or inflammation in the treatment site.

(36) In some embodiments, the compressive layer **34** can include a first plurality of voids (e.g., through holes) and an optional second plurality of voids extending between the first surface **70** and the second surface **74**. The first plurality of voids and the second plurality of voids can be used with compressive layer **34** made of the textile material **118**, the foam material **146** or other materials. The first plurality of voids are positioned proximate the ends **90**, **94** of the first elongated portion **78** and the ends **98**, **102** of the second elongated portion **82** The first plurality of voids are oriented so that the first plurality of voids open in a direction that is generally parallel to the vertical axis **50**. Accordingly, in the illustrated embodiment, the compression in the vertical direction is based on the pores and not the first plurality of voids. The first plurality of voids are elongate in the generally longitudinal direction and have thicknesses oriented generally in the lateral direction. Accordingly, in the presence of negative pressure, the first plurality of voids are configured to collapse laterally towards (e.g. perpendicularly with respect to) the longitudinal axis **42** and in the vertical direction defined by the vertical axis **50**. The first plurality of voids can be larger than the second plurality of voids to generate more lateral compression proximate the ends **90**, **94** of the first elongated portion **78** and the ends **98**, **102** of the second elongated portion **82**.

(37) The second plurality of voids can be positioned in the central portion **106** of the compressive layer **34**. The second plurality of voids are shaped and/or oriented to provide less compression than the first plurality of voids. For example, in some embodiments, the second plurality of voids could be smaller than the first plurality of voids. In other embodiments, the adjacent voids of second plurality of voids could be spaced further apart than adjacent voids of the first plurality of voids.

Non-Adherent Layer

(38) Referring again to FIG. **3**, the non-adherent layer **38** is shown to include a first surface **174** and a second, wound-facing surface **178** opposite the first surface **174**. When the negative pressure dressing **10** is applied to the treatment site **14**, the first surface **174** faces away from the treatment site, whereas the second surface **178** faces toward the treatment site. In some embodiments, the first surface **174** of the non-adherent layer **38** contacts the second surface **178** of the compressive layer **34**. In some embodiments, the second surface **178** of the non-adherent layer **38** contacts the surface of the patient.

(39) The non-adherent layer **38** is made of a material that is fluid-permeable and intended to not irritate the patient's skin. In the illustrated embodiment, the non-adherent layer is a polyester pique-knit textile material, such as Milliken Textile material. In other embodiments, other permeable and non-irritating textile materials can be used. The non-adherent layer **38** can also be treated with antimicrobial materials. In the illustrated embodiment, the non-adherent layer **38** includes silver ions as an antimicrobial material. Other anti-microbial materials may be used in other embodiments.

Deployment of the Treatment Area Therapy System

(40) Referring now to FIGS. **7** and **8**, the negative pressure dressing **10** is shown engaged with the treatment site **14** of a patient to provide NPT to the treatment site **14**. The treatment site **14** includes a joint **182**, a first portion **186** of the surface of the patient adjacent and generally proximal to the joint **182**, and a second portion **190** of the surface of the patient adjacent and generally distal to the joint **182**. In the illustrated embodiment, the joint **182** is a knee joint. In other embodiments, the joint **182** can be another type of joint, such as an ankle joint, a hip joint, a wrist joint, an elbow

joint, or a shoulder joint.
(41) The negative pressure dressing **10** is in fluid communication with the NPT system **12**. The NPT system **12** includes a negative pressure source **194**, such as a pump, and negative pressure conduit **198** in fluid communication with the negative pressure source **194**. The negative pressure dressing **10** is positioned over the treatment site **14** such that the connecting portion **26** is positioned over the knee joint **182**. The first elongated portion **78** is secured to the first portion **186** of the surface of the patient. The second elongated portion **82** is secured to the second portion **190** of the surface of the patient. The adhesive of the margin **62** forms fluid-tight seal around a perimeter of the margin **62**.
(42) The negative pressure conduit **198** is then engaged with the negative pressure port **58** formed in the sealing layer **30**. The negative pressure source **194** is actuated to generate negative pressure within the negative pressure dressing **10**. In response to the negative pressure provided by the negative pressure source **194**, the compressive layer **34** first contracts in the generally vertical direction and then contracts in the generally lateral direction. The plurality of channels **114** adjacent each rib of the plurality of ribs **110** facilitates a combination of generally lateral contraction and upward bending proximate the ends **90**, **94** of the first elongated portion **18** and the ends of the second elongated portion **22**. This lateral contraction and bending generates a lifting force about a perimeter of the knee joint **182**, which facilitates circulation and/or lymph flow proximate the knee joint **182**, thereby reducing swelling and/or inflammation proximate the knee joint **182**.
(43) FIG. **7** illustrates the negative pressure dressing **10** secured to a bent knee joint **182**. FIG. **8** illustrates the negative pressure dressing **10** secured to a straight knee joint **182**. As shown in FIG. **7**, the plurality of articulations **66** formed in the sealing layer **30** splay apart to facilitate bending of the knee joint **182**. As is shown in FIG. **8**, the plurality of articulations **66** are not splayed apart, allowing the negative pressure dressing **10** to lie against the straight knee joint **182**.
Configuration of Exemplary Embodiments
(44) The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements can be reversed or otherwise varied and the nature or number of discrete elements or positions can be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. As used herein, the term “approximately” may be substituted with “within a percentage of” what is specified, where the percentage includes 0, 1, 4, and 10 percent. The order or sequence of any process or method steps can be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions can be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

Claims

1. A compressive layer for treating a treatment area proximate to a joint with negative pressure, the compressive layer comprising: a first elongated portion including a first end and a second end; a second elongated portion spaced from the first elongated portion and including a first end and a second end; a central portion extending between the first elongated portion and the second elongated portion; and a plurality of channels formed in at least one of the first end and the second end of the first elongated portion and at least one of the first end and the second end of the second elongated portion; and a plurality of ribs formed between adjacent channels of the plurality of channels, wherein a width of adjacent ribs of the plurality of ribs is graduated in a laterally inward direction.

2. The compressive layer of claim 1, wherein the first elongated portion and the second elongated portion are configured to be positioned proximate to the joint.
3. The compressive layer of claim 1, wherein the central portion is configured to overlie at least a portion of the joint.
4. The compressive layer of claim 1, wherein the plurality of channels have a depth less than a thickness of the compressive layer.
5. The compressive layer of claim 4, wherein the thickness of the compressive layer extends between a first surface and a second surface of the compressive layer, wherein the depth of the channels extend from the first surface toward the second surface of the compressive layer, and wherein the second surface is configured to face the treatment area.
6. The compressive layer of claim 1, wherein compression of the compressive layer generates a lifting force.
7. The compressive layer of claim 1, wherein the width of the adjacent ribs of the plurality of ribs decreases in the laterally inward direction.
8. The compressive layer of claim 1, wherein the plurality of channels is a first plurality of channels, and wherein the compressive layer further comprises a second plurality of channels formed in the other of the first end and the second end of the first elongated portion and the other of the first end and the second end of the second elongated portion.
9. The compressive layer of claim 8, wherein compression of the compressive layer proximate the first plurality of channels and the second plurality of channels generates a lifting force configured to surround a circumference of the treatment area.
10. The compressive layer of claim 1, wherein the compressive layer is configured for compression in at least a vertical direction and at least a horizontal direction, the compression in the horizontal direction being larger than the compression in the vertical direction so that the compressive layer compresses more in the horizontal direction than in the vertical direction.
11. The compressive layer of claim 10, wherein the plurality of channels are configured to facilitate compression in the horizontal direction.
12. The compressive layer of claim 1, wherein the compressive layer comprises a textile material including a knit pattern, the plurality of channels being formed by the knit pattern.
13. The compressive layer of claim 1, wherein the compressive layer comprises a foam material, the plurality of channels being formed by a plurality of welds in the foam material.
14. The compressive layer of claim 1, wherein the plurality of channels are configured to distribute negative pressure and/or fluid through the compressive layer.
15. The compressive layer of claim 5, wherein the second surface of the compressive layer has a first compression modulus in a horizontal direction and a second compression modulus in a vertical direction, the second compression modulus being larger than the first compression modulus so that the second surface of the compressive layer compresses more in the horizontal direction than in the vertical direction.
16. The compressive layer of claim 5, wherein the first surface of the compressive layer has a first compression modulus in a horizontal direction and a second compression modulus in a vertical direction, the second compression modulus being larger than the first compression modulus so that the first surface of the compressive layer compresses more in the horizontal direction than in the vertical direction.